



JCLR 2022 | Bringing BERT to Vision: BEiT Pre-training Image Transformers by Predicting Visual Tokens

Harbin Institute of Technology and Microsoft Research propose masked image modeling (MIM) to pre-train ViTs with no labels

Visual Transformers (ViT) have an awkward habit: they are strong, but hungry. To train well they usually want large amounts of labeled supervision. In natural language processing, BERT long ago pointed to a different path — mask out some of the words in a sentence, ask the model to fill them back in, and it learns general-purpose representations from oceans of unlabeled text. The obvious question is whether images can be pre-trained the same way.

Porting BERT's masked language modeling straight onto pixels does not work cleanly. Language comes with a ready-made vocabulary, so every masked position has a well-defined discrete answer. Images have no such vocabulary, and if you ask the model to regress the raw masked pixels, it spends much of its capacity on high-frequency texture instead of the semantics you actually care about.

BEiT, from Harbin Institute of Technology and Microsoft Research (JCLR 2022), answers with a simple twist: do not predict pixels — predict visual words. BEiT introduces masked image modeling (MIM). It first uses a discrete image tokenizer to turn an image into discrete visual tokens, then masks a portion of the image patches and trains a ViT to predict the visual tokens at the masked positions from the visible ones. None of the pre-training uses any human annotation.

The overview figure draws BEiT's two "views" side by side; read it left to right. The original image on the left is both cut into image patches and encoded by the image tokenizer into a grid of discrete visual tokens. During pre-training, roughly 40% of the patches (the gray patches) are masked and replaced with a special mask embedding [M], the corrupted sequence is fed to the BEiT Encoder, and the objective is to predict the original image's visual tokens at the masked positions.

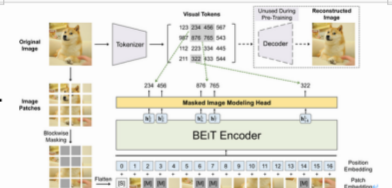


Figure 1. Overview of BEiT pre-training. Before pre-training, an image tokenizer is learned via autoencoding-style reconstruction, turning an image into discrete visual tokens. During pre-training each image has two views — image patches and visual tokens; some patches are masked and replaced with [M], and the backbone ViT predicts the original visual tokens from the completed image. The decoder on the tokenizer's right is only used to learn the vocabulary and is unused during pre-training.

Paper: <https://arxiv.org/abs/2106.08254> ; Code: <https://github.com/microsoft/unilm/tree/master/beit>

Why predict tokens instead of pixels

This is the key departure from earlier self-supervised vision methods. Contrastive approaches lean on carefully designed augmentations and negatives, while per-pixel regression tends to sink into detail. Discrete visual tokens compress the prediction target into a classification problem that keeps the semantics and mirrors BERT's "predict the masked word"



预测目标可以来自白地知识：在整个数据集上，对随机位置的位置，最大化被遮盖位置上正确视觉 token 的数。训练完成后，只需在 ViT 上接一个任务并调整模型，就能迁移到下游任务。作者还指出，配合小范围初始化和速度增益，这套训练在大模型上也能收敛稳定：从理论上讲，MIM 可以被理解为一种变分自编码器的其中一个阶段。

结果：无标注预训练，逼平甚至超过有标注

在 ImageNet-1K (约 1.2M 张无标注图像) 上自监督预训练、再微调之后，BEiT 基线版 (BEiT-B) 达到 83.2% 的 top-1 准确率，大模型版 (BEiT-L) 达到 85.2%，在 384x384 分辨率下进一步提升到 86.3%，这成绩不仅超过从零训练的同等模型，也优于 DINO、MoCo v3 等自监督对比学习基线。更有趣的是，仅在 ImageNet-1K 上预训练的 BEiT-B4，在分辨率下甚至超过了更大的 ImageNet-22K 上带有标注预训练的 ViT-B4。

语义分割上同样表现，在 ADE20K 上，BEiT 取得 45.6 的 mIoU，若先在 ImageNet 上跑一轮中间微调再迁移到 ADE20K，mIoU 可进一步提升到 47.7，这说明 MIM 学到的表示不只对分类有用，对需要密集预测的任务也能迁移。

洞察：真正起作用的是「预测离散 token」

作者用一些消耗把关键信息逐一拆开。把 MIM 的预测目标从「离散视觉 token」换成「还原被遮盖」，ImageNet 微调准确率从 82.86% 掉到 81.04%，ADE20K 的 mIoU 从 44.65 掉到 41.38 — 预测 token 才是核心配方。去掉快速道路 (改成随机噪声块) 再训练 (82.77% / 42.83)，而把预测目标拉长到 800 块，结果能提升到 83.19% / 45.58，说明这个目标能帮算力还能继续吃到的收益。

表 1: BEiT 关键结果 (ImageNet-1K 预训练/微调 / ADE20K mIoU)

设置	ImageNet top-1	ADE20K mIoU
预测视觉 token (BEiT)	82.86%	44.65%
改为还原遮盖	81.04%	41.38%
去掉快速道路	82.77%	42.83%
预测 800 块	83.19%	45.58%

一个额外实验验证 MIM 的视觉信息。作者在预训练好的 BEiT-L 上，取不同参考点查看由注意力图。尽管整个预训练没有用到任何人工标注，注意力图能把参考点所在的物体从背景中「圈」出来。

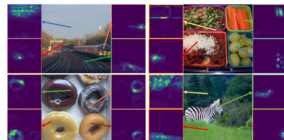


图 2 不同参考点的注意力图。BEiT 的注意力图能够定位到人工标注的情况下分类图像中的物体，说明 MIM 预训练已经学到了带有物体语义的表示。

这从侧面说明，预测视觉 token 这个看似「薄格式」的目标，实际上迫使模型去理解图像的语义结构，而不仅仅是记忆纹理。

小结与边界

BEiT 的贡献可以用一句话概括：预测 token，而不是像素。通过预训练图像建模与离散视觉 token，它把 BERT 式的自监督预训练成功带到了视觉 Transformer，在分类与分割上逼平甚至超过了有标注与对比学习的预训练方案。对实践者而言，这意味着可以先用大量无标注图像预训练一个骨干网，再用少量标注做下游微调。

也要看清边界：BEiT 依赖一个外置预训练的图像分词器 (来自 DALL-E 的 vPAE) 来提供预训练目标，词表质量会影响上游；论文的主要结论建立在 ImageNet-1K 预训练与 ImageNet-ADE20K 评测之上，即便如此，BEiT 仍被作为计算机视觉的一个「BERT 时刻」——它证明了 BERT 式预训练不只属于文本，对图像同样有效，而且效果很好。